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Influence of Ga incorporation on photoinduced phenomena in Ge–S based glasses

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To study the influence of Ga addition on photoinduced effect, GaGeS glasses with constant atomic ratio S/Ge = 2.6 have been prepared. Using Raman spectroscopy, we have reported the effect of Ga on the structural behavior of these glasses. An increase of the glass transition temperature T_g, the linear refractive index and the density have been observed with increasing gallium content. The photoinduced phenomena have been examined through the influence of time exposure and power density, when exposed to above light bandgap (3.53 eV). The correlation between photoinduced phenomena and Ga content in such glasses are shown hereby.

1. Introduction

Chalcogenide glasses are attractive materials because they combine relatively low transmission losses with the unique photoinduced phenomenon [1]. These glasses, when illuminated by bandgap light, usually exhibit a blue or a red shift of the optical gap, namely photobleaching (PB) or photodarkening (PD), respectively. Furthermore, such photoinduced changes can be in general irreversible, i.e. the changes are permanent after irradiation, or reversible in which case the changes can be removed by annealing at the glass transition temperature, T_g.

In particular, Ge–S glass systems are of continuous interest in terms of the photoinduced changes that present after illumination with light. Moreover, a considerable change of the optical parameters and photoinduced properties are observed by introducing specific metals. For example, the inclusion of Bi or In into unexposed Ge–S films leads to PD, increase in refractive index n, and a considerable decrease in E_g. This effect is associated to the increase of the disorder and creation of new localized sates near the valence band [2–4]. Otherwise, the Pb incorporation in the Ge–S system [5–7] eliminates PD. In this case, the reason for the disappearance of the PD results from the disappearance of the photostructural change.

In our previous investigations of the photoinduced changes in Ga containing glasses, we reported PB associated with photoexpansion (PE) effect [8]. Moreover, depending on the time exposure, we found that this PE effect was irreversible [9]. Studies on local structure of glasses in the Ge–Ga–S system have been already investigated [10], but none has investigated the influence of metals (Ga in our case) on the photoinduced phenomena of the glasses. In the present paper, we report the effects of Ga addition in the Ge–S glasses. Results of the optical properties of these glasses and a correlation between the glass structure and its photoinduced properties are presented.

2. Experimental

High-purity GeS, Ga_2S_3 and S raw materials (5 N, from Asteran Ltd, Russia) were weighed in 10 g into fused quartz ampoules in a N_2 gas-filled glove box with <1 ppm H_2O and O_2 concentrations. The fused quartz ampoules were washed in advance with distilled water, soaked for 3 min in 25% HF acid, re-washed with distilled water and dried at 500 °C in an oven. The ampoules containing the raw materials were then sealed under vacuum of about 10^{-3} mbar. The batches were slowly heated up to 950 °C and maintained at this temperature for 10 h in order to permit thorough reaction of the raw materials and to avoid explosion due to the high vapor pressure of sulfide. Then, the melts were cooled in air for a few seconds and the samples obtained were annealed at the glass transition temperature. Finally the glass rod was removed from the ampoule, cut into slices and polished so as to obtain 2 mm thickness samples.

We progressively introduce Ga in $Ge_{28}S_{72}$ keeping the atomic ratio S/Ge = 2.6 constant, to elucidate the role of Ga in similar backbone structure.

Characteristic temperatures (T_g for glass transition temperature, T_x for onset of crystallization) were determined by differential scanning calorimeter (DSC) using a TA2910 Instrument equipment. The estimated error on the temperature is 2 °C for glass transition

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and onset of crystallization, which are obtained from tangents intersection. Powdered samples were set in aluminum pans and heated under N_2 atmosphere at 10 °C/min.

The density (ρ) was determined by Helium (Micromeritics AccuPycc 1330) picnometer. The volume of each sample was measured at least 10 times, and the density calculated from their average. The refractive index was measured at 632.8 nm by a M-lines apparatus (Metricon model 2010) based on the prism coupling technique [11].

The UV–vis absorption spectra were recorded between 300 and 700 nm at room temperature using a Cary 500 spectrophotometer (Varian).

Raman spectroscopy was carried out using a Renishaw microscope equipped with a He–Ne laser with a wavelength of 633 nm. The power of the excitation beam used for the measurements was about 5 mW, which is too low to affect the glass surface. The laser beam was focused on the sample surface by a 100× microscope objective and then recorded from 100 to 600 cm^{-1} with a resolution of 2 cm^{-1}.

To investigate the photoinduced effect, glass samples were exposed to ultraviolet (UV) light at 351 nm from an Ar^+ ion laser (from Innova 308C) for different times (0–180 min) and power densities (0.25–0.75 W/cm^2). All experiments were performed at room temperature.

3. Results

3.1. Glass properties

The characteristic temperatures of the glasses obtained by DSC are reported in the Table 1. We have also included in this table the value of ΔT (=T_x−T_g) generally used as an estimate factor of glass stability against devitrification.

It can be seen from Table 1 that T_g value increases from 301 °C to 349 °C as the amount of Ga rises from 0% to 15%. Besides, a decrease of ΔT from 269 °C to 48 °C is observed when Ga concentration is varying from 0% to 15%. This indicates that glasses with higher amount of Ga present a higher tendency to devitrificate.

The density (ρ) and linear refractive index (n) increase as the amount of Ga increases, varying from 2.670 to 3.155 and 2.000 to 2.206, respectively.

The values of the optical bandgap, E_g, of the glasses were calculated from the $\alpha = 0$ intersects of $(\alpha h\nu)^{1/2}$ versus hν plots, where α is the absorption coefficient estimated according to [12], and hν is the photon energy. The error in determination of E_g values is evaluated as ±0.01 eV.

3.2. Structural studies – Raman scattering

Fig. 1 presents Raman spectra performed in the range spectrum 100–600 cm^{-1}. We confirmed that all samples were amorphous, as the Raman spectra exhibited no sharp crystalline peak. These spectra are in good agreement with the spectra of Ge–Ga–S glasses reported earlier by others authors [10,13].

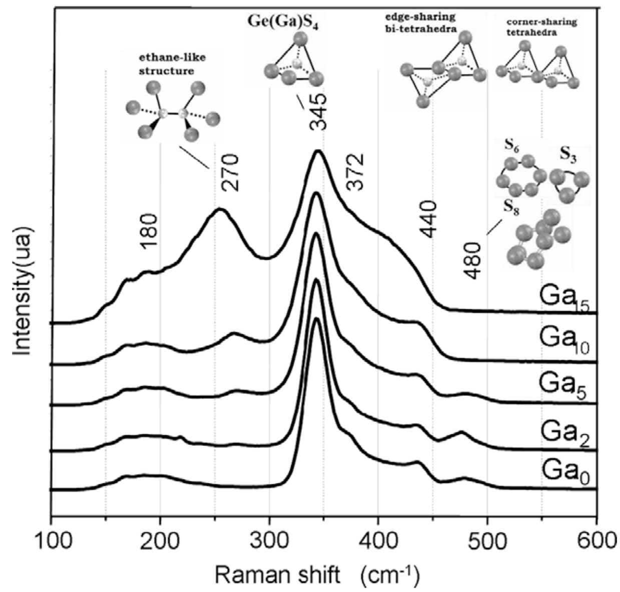

Fig. 1. Reduced normalized Raman spectra of GaGeS glasses with the corresponding structures.

The basic structure of GeS_2 glass is formed by $GeS_{4/2}$ tetrahedra which are connected through a bridging sulfur to form a three dimensional network. The strongest band at 345 cm^{-1} is a Ge–S arising from symmetric stretching of [$GeS_{4/2}$] units. In addition, two different ways of connecting the tetrahedra provide two additional bands. Firstly, the band at 372 cm^{-1} is known as a companion band (A_c) which is due to the vibration of two edge-shared tetrahedra [14]. Secondly, the band at 440 cm^{-1} is due to the vibration of two tetrahedra connected through a bridging sulfur at the corner [15]. The origin of the band at 180 cm^{-1} is not totally clear. In accordance with Lucovsky et al. [16], this band may be correlated to asymmetrical bending (ν_4) vibration of XY_4 tetrahedral molecules. The band at 270 cm^{-1} is due to the vibration of two tetrahedra with a S_3Ge(Ga)–(Ga)GeS_3 ethane-like structure, as it is represented in Fig. 1. Finally, the band observed at 480 cm^{-1} arises from the stretching of S–S bonds in short S_n chains or S_8 rings.

Addition of Ga into $GeS_{2.6}$ has resulted in pronounced changes in the Raman spectra as illustrated in Fig. 1. As the concentration of Ga increases, the bands at 180, 270, 372 and 440 cm^{-1} increase and a shift to lower frequency is observed for the 270 and 440 cm^{-1} bands. On the other hand, the intensity of the band at 480 cm^{-1} decreases progressively with increasing Ga content and totally disappears for glasses with Ga content above 10% (in mol%).

3.3. Photoexpansion studies – evolution of thickness

The irradiation experiments have been performed on samples of 2 mm thickness using a UV laser with an energy (E = 3.53 eV) above the optical bandgap of all the glasses (cf. Table 1). The

Table 1
Characteristic temperatures, linear refractive index n, density ρ, optical bandgap E_g, absorption coefficient $\alpha_{351\ nm}$ and penetration depth $\delta_{351\ nm}$ at the illuminating wavelength (λ = 351 nm) of the GaGeS samples.

Sample	Composition			T_g (±2 °C)	T_x (±2 °C)	ΔT (±4 °C)	n (±0.001)	ρ (g/cm^3)	E_g (eV)	α_{351nm} (cm^{-1})	δ_{351nm} (μm)
	Ga	Ge	S								
Ga0	0	28	72	301	570	269	2.000	2.670	2.61	69	145
Ga2	2	27	71	307	563	256	2.102	2.779	2.57	61	164
Ga5	5	26	69	315	520	205	2.152	2.790	2.63	72	139
Ga10	10	25	65	352	446	94	2.165	2.868	2.71	63	159
Ga13	13	24	63	350	420	70	2.183	2.908	2.33	–	–
Ga15	15	24	61	349	397	48	2.206	3.155	2.12	71	141

absorption coefficient $\alpha_{351\mathrm{nm}}$ at the illuminating wavelength, $\lambda = 351$ nm, has been calculated for each sample from the UV–vis transmission data. The obtained values are reported in Table 1 with the corresponding penetration depth $\delta_{351\mathrm{nm}}$, calculated from $\delta = \alpha^{-1}$. Profile measurement for GaGeS glasses were carried out as a function of the applied power density and exposure time. All the experiments were performed using a copper mask as illustrated in Fig. 2(a). The Fig. 2(b) presents a 3D scan of the surface of the Ga10 glass realized by perfilometry after irradiations with different power densities and times through the mask.

The values of thickness change of the exposed areas are reported in Fig. 3 in function of power densities and exposure durations for the different GaGeS glasses.

Firstly, it can be noticed that results for the Ga0 sample were not reported here since no expansion was observed.

The Ga2 and Ga15 samples present lower values of expansion (maxima of $\sim$ 300 nm) obtained with the higher power density (0.75 W/cm^2). They also present a similar behavior according exposure time with a maximum of expansion reached after 120 min.

For glasses with Ga concentration varying from 5% to 10%, it is observed an exposure time dependency of the photoexpansion. Moreover it can be noted the existence in both cases of an optimal power density (0.42 W/cm^2) to get the larger expansion values after 180 min of exposure.

The highest values of expansion have been obtained with the Ga10 sample with a maximum measured expansion of about 700 nm after 180 min of exposure at 0.42 W/cm^2.

3.4. Photoinduced effects – study of the reversibility

The Ga10 sample was chosen to evaluate the reversibility of the photoexpansion phenomenon. The same glass slice exposed at different power densities during different times, used for the above study of expansion, was submitted to an heat treatment at 350 °C (near T_g) during 40 min. Photographs obtained by optical microscopy of the exposed areas after heat treatment are presented for three expansion thicknesses: 190, 380 and 600 nm in Fig. 4(a), (b) and (c), respectively.

We can note the tendency toward formation of cracks resulting from the internal stress associated with the photoexpansion, i.e. the tendency to cracking increases with the thickness of the expanded area. The shape of the crack indicates that this region has

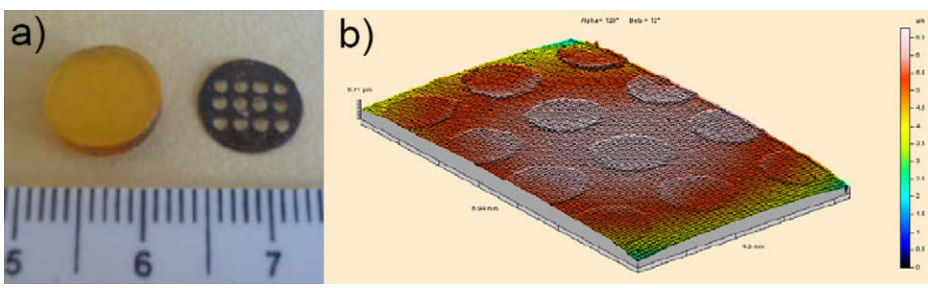

Fig. 2. (a) Photography of the Ga10 sample and the mask used for exposure experiments. (b) 3D surface of the Ga10 sample irradiated at different power densities and exposure times, obtained by perfilometry.

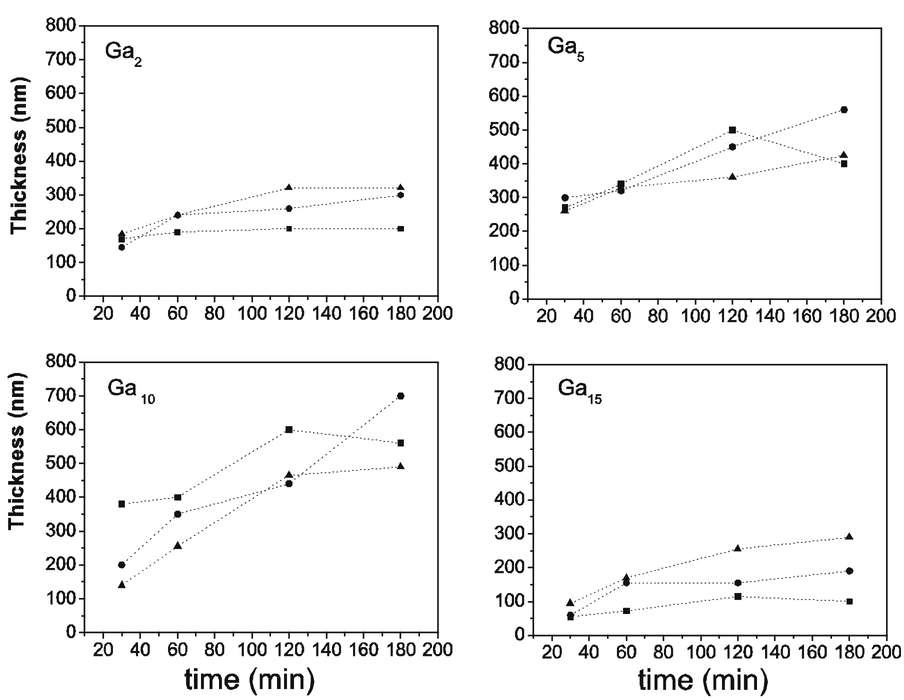

Fig. 3. Evolution of thickness of exposed area for Ga2, Ga5, Ga10 and Ga15. Laser irradiation power density (■) 0.25 W/cm^2, (●) 0.42 W/cm^2, and (▲) 0.75 W/cm^2. Lines are drawn as guide for the eye.

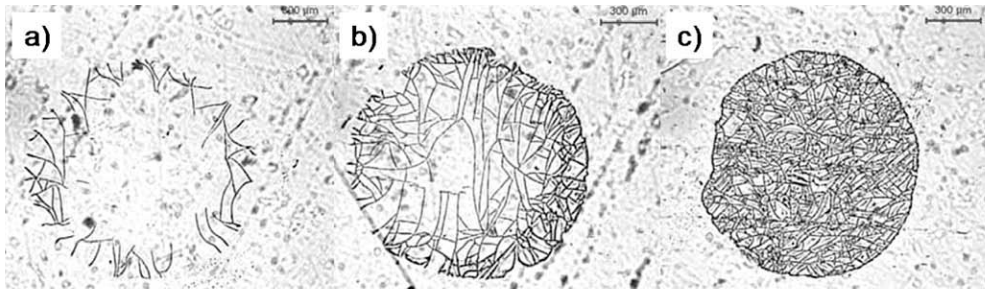

Fig. 4. Microphotographies of three exposed areas on the Ga10 sample of (a) 190 nm, (b) 380 nm and (c) 600 nm thickness, after heat treatment at 350 °C during 40 min.

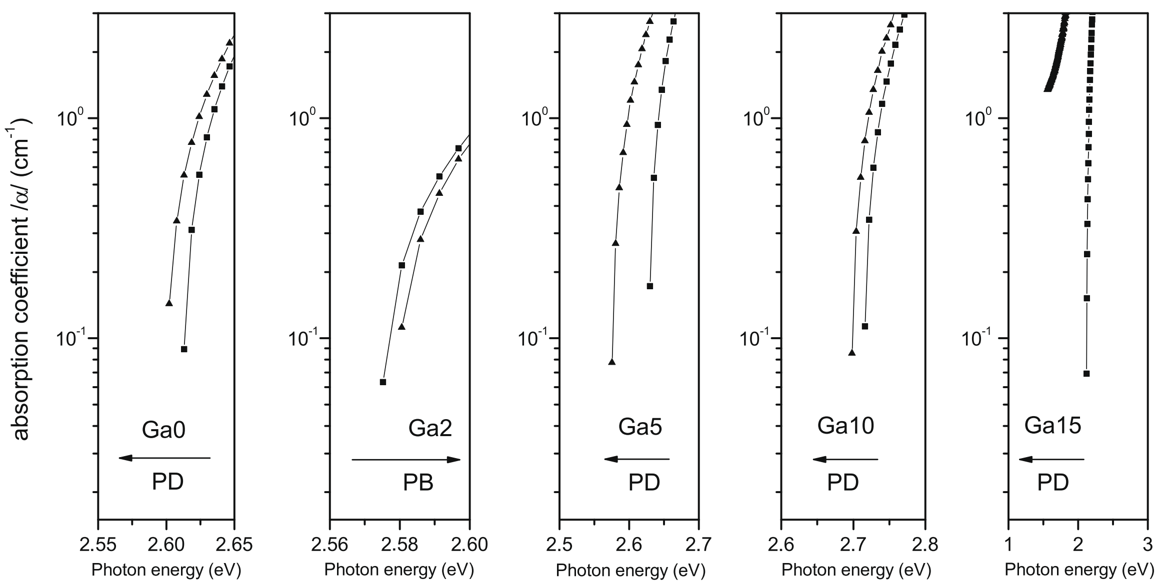

Fig. 5. Optical absorption spectra for Ga0, Ga2, Ga5, Ga10, and Ga15 before (■) and after (▲) exposure at 0.42 W/cm^2 during 5 h.

been submitted to stress resulting from the different power density. One expects large shear stresses at the interface between expanding and non-expanding regions.

3.5. Photodarkening and photobleaching effects

In order to facilitate the optical absorption measurements, the GaGeS samples have been irradiated on a larger area. A divergent lens has been used to obtain a 4 mm diameter spot on the sample. The laser beam power was then rectified to reach an incident power density of 0.42 W/cm^2 at the surface of the glass. The incident beam power density was checked with a powermeter before starting each exposure.

Fig. 5 shows the optical absorption spectra for the GaGeS glasses before irradiation and after 5 h exposure. Photodarkening and photobleaching was observed by comparison between the optical bandgaps before and after illumination.

A photodarkening effect was observed after irradiation for all the samples (including for the Ga-free glass), less the Ga2 sample which presented a photobleaching effect.

4. Discussion

Table 1 shows the compositional dependences of some properties in the prepared GaGeS glasses. The observed increase in density is assigned to the fact that the GaS_4 tetrahedra are more efficiently packed than GeS_4 tetrahedra [17] which compensates the effect of the slightly atomic mass on the whole. The monotonic increase in refractive index with increasing Ga content is expected due the increase packing density and the larger polarizability of Ga^{3+} ions in comparison with Ge^{4+} ions. The increase of T_g can be related to the structural changes in the vitreous network by Ga addition.

The Raman data of the glass samples show the formation of GaS_4 tetrahedra from Ga_2S_3, resulting in a deficiency of sulfur in the glasses since the S/Ga ratio of Ga_2S_3 is 1.5, which is less than 2 required for forming tetrahedra. The band at 270 cm^{-1} increased in amplitude as the amount of Ga increased. This behavior is mainly due to the formation of $S_3Ge(Ga)$–$(Ga)GeS_3$ ethane-like structure and to the removal of a sulfur originally connecting two tetrahedra to provide an extra sulfur needed for the formation of GaS_4 tetrahedra.

The strong band at 345 cm^{-1} is a Ge–S vibration arising from bond bending modes and the A1 type bond stretching mode of $GeS_{4/2}$ tetrahedra [18]. In addition, Ga_2S_3 forms GaS_4 tetrahedra and participates in the formation of a cross-linked structure of the glasses. Since the atomic mass of Ga is similar to that of Ge, the frequency of the GaS_4 vibration is only slightly larger than that of the GeS_4 tetrahedra [10]. A resolution between Ge–Ge and Ga–Ga bonds is not possible from the present Raman spectra because of the similarity in the atomic masses of these two elements.

The band observed at 440 cm^{-1} is currently assigned to vibrational modes of edge–sharing bi-tetrahedra, $Ge_2S_{2+4/2}$. This structure is known to exist in crystalline GeS_2, and its existence has also been demonstrated in a glassy phase [19]. On the other hand, some researchers have assigned the 440 cm^{-1} band to the S–S stretching mode, coming from the dimmers at the edges of a molecular cluster [20].

The last modification is the decrease in amplitude of the band at 480 cm^{-1}, which arises from the stretching A1 mode of S–S bonds of short S_n chains. The progressive addition of Ga to $GeS_{2.6}$ results in a defect of sulfurs atoms, which prevents the maintenance of α-GeS_2-type tetrahedral units linkage. In the series studied in this work, the Ga10 sample is a stoichiometric composition corresponding to $(Ga_2S_3)_{16.67}$–$(GeS_2)_{83.33}$. So for the richer Ga content glasses, the sulfur defect implies a compensation by the formation of edge linkages between $(Ge/Ga)S_4$ tetrahedral units, as it is shown in the Raman data with the vanishing of S_n chains associated band.

Previous structural study by EXAFS technique [21,22] has shown that the addition of Ga gives rise to GaS_4 tetrahedral units with Ga–S distances significantly larger than Ge–S distances in GeS_4 tetrahedral units. The Ga insertion in the structural organization may maintain the global α-GeS_2 structure, but distorting it through edge or corner-sharing linkage of GaS_4 tetrahedral substituted for GeS_4 units.

Regarding the photoexpansion effect observed on GaGeS bulk glasses, a systematic study was done to evaluate the effect of Ga addition. As shown in Fig. 3, all the curves that correspond to the thickness of the expanded area increase depending on the Ga concentration. No expansion effect was observed for the Ga-free sample (Ga0) while it has been observed for all the compositions with $x > 0$. Higher values have been reached for $5 \leqslant x \leqslant 10$ samples, with a maximum of 700 nm in thickness for the Ga10 sample with 0.42 mW/cm^2 power density. For higher Ga concentration (Ga > 10), the magnitude of expansion decreases. Incidentally, the present observation of higher photoexpansion by using above bandgap illumination, is rather opposite to the one observed for As_2S_3. A reversible effect of thickness increase (0.5%) in As_2S_3 films induced by bandgap illumination was shown in one of the first related reports [2]. Later, a 'giant' (about 5%) photoexpansion (PE) has been obtained with subbandgap intense CW laser light [3]. Loeffler et al. [23] in their study on the effect of illumination on similar chalcogenide glasses ($Ge_{25}Ga_5S_{70}$) observed a $\Delta V/V = 0.004\%$. They have proposed a microscopic model where heteropolar bounds are broken by absorption of light energy photons and new homopolar bounds are formed simultaneously. In this case the interatomic distance between the new bonds formation (Ge–Ge) and (S–S) do not explain such high photoexpansion.

Therefore in our previous work [24], we have suggested the dependence of photoexpansion with the exposure atmosphere. We have proposed that the action of the light is to weaken and break bound and that oxygen from atmosphere induce a surface oxidation resulting in the increase in the volume. In fact, when Ga is added to Ge–S system, it is substituted into the glass matrix, resulting in one electron is missing from one of the four covalent bonds Ge–S glass. Thus, the gallium atom can accept an electron from a neighboring atoms (in our case, oxygen) to complete the fourth bond. During irradiation, this gallium defect implies a compensation by the formation of edge linkages between (Ge/Ga)–O units inducing a photoexpansion effect. This fact is confirmed since there is no photoexpansion in Ge–S glass.

The change of the optical absorption edge as the Ga content increases (Fig. 5) could be related to preparation conditions leading to defects created after illumination. Fritzsche [25] and Ganjoo et al. [26] proposed that photodarkening can generally be assigned into either one of two categories: they are based on localized changes in the bonding configuration at specific lattice sites, or they invoke larger-scale rearrangements of the whole network. In our case, Ga addition may cause a rearrangement of the whole Ge–S glass network as shown by Raman scattering data. However, further experiments with different amounts of Ga are under study in order to elucidate these optical absorption changes.

Generally photobleaching is observed in Ge–S glasses [27], but in our case a photodarkening has been observed for the Ga0 sample after irradiating. Experiments performed on $Ge_{1-x}S_x$ glasses (with x = 62, 66, 69, 72 and 75) have shown both photobleaching and photodarkening after illuminating at 351 nm, depending on the sulfur amount in the glass composition. For low concentration of sulfur ($x \leqslant 69\%$), a photobleaching is observed while above this concentration a photodarkening occured. Nevertheless, the mechanism remains unclear and is still under study.

5. Conclusion

To evaluate the influence of Ga incorporation on photoinduced phenomena in Ge–S based glasses, samples with constant atomic ratio S/Ge = 2.6 have been prepared with different amounts of Ga. An increase of glass transition temperature T_g, linear refractive index and density has been observed as the amount of Ga increases while the stability against crystallization decreases. Raman spectroscopy data indicate that the progressive addition of Ga in the Ge–S system results in a sulfur deficient glassy network, promoting the formation of homopolar metal–metal bonds and edge-shared $Ge(Ga)S_{4/2}$ tetrahedra.

We have demonstrated that the presence of Ga into the Ge–S binary glass system is a requirement for photoexpansion effect, because this addition promotes a 'hole' in the glass matrix, leading to Ga(Ge)–O bonds formation. Therefore, the maximum value of photoexpansion is reached for 10% of Ga content. Furthermore, annealing experiments at glass transition temperature performed on the exposed samples have shown the irreversibility of the photostructural changes.

Regarding the optical bandgap photoinduced changes, further studies are still in progress to have a better understanding of the observed phenomenon.

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